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SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405
Faculty : Dr. Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to realworld problems. (BTL-3)

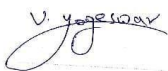
Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I **V Yogeswar / 192425175** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



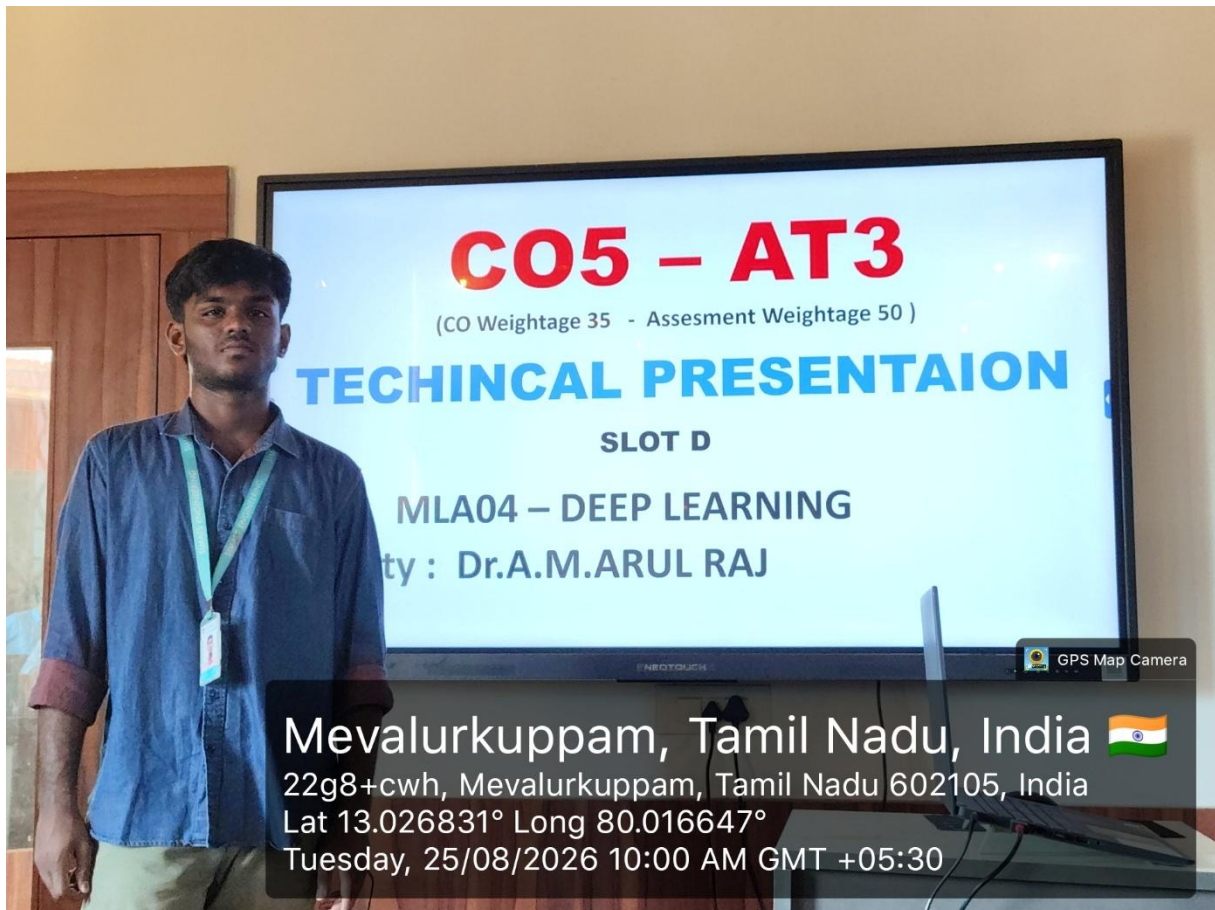
Name of the student: V Yogeswar

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS: 3. Deep Belief Networks and Boltzmann Machines

Develop a technical presentation comparing Deep Belief Networks and Boltzmann Machines. Explain their architecture, energy-based learning, hidden and visible units, training process, and feature-learning capabilities. Illustrate their use in a suitable real-world application.

PRESENTATION ANSWER:

1. Introduction

- Boltzmann Machines (BM) and Deep Belief Networks (DBN) are neural networks used for unsupervised learning.
- They learn hidden patterns and useful representations directly from input data.
- Both are based on the concept of energy-based learning.
- DBNs can be considered a deep extension built using Restricted Boltzmann Machines (RBMs).

2. Boltzmann Machine Architecture

A Boltzmann Machine contains:

- Visible units: Receive the input data.
- Hidden units: Learn underlying features from the input.
- Connections exist between units, allowing the network to model relationships.

Basic structure:

Hidden Units

H1 H2 H3

\ | /

\ | /

V1--V2--V3

Visible Units

- In a traditional Boltzmann Machine, connections can exist between visible-visible, hidden-hidden, and visible-hidden units.
- The network is generally stochastic, meaning unit states are determined probabilistically.

3. Deep Belief Network Architecture

A DBN consists of multiple layers of hidden units.

Hidden Layer 3

H31 H32 H33

↓

Hidden Layer 2

H21 H22 H23

↓

Hidden Layer 1

H11 H12 H13

↓

Visible Layer

V1 V2 V3

- The bottom layer receives the input.
- Each hidden layer learns increasingly complex features.

- DBNs are commonly constructed by stacking RBMs.
- The upper layers capture more abstract representations.

4. Energy-Based Learning

Both models use an energy function.

The main idea is:

- Configurations with low energy are considered more likely.
- Configurations with high energy are considered less likely.
- During training, the model adjusts weights so that observed data receives lower energy.

A simplified Boltzmann energy function is:

$$E(v, h) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_{i,j} v_i w_{ij} h_j$$

Where:

- v_i = visible unit
- h_j = hidden unit
- a_i, b_j = biases
- w_{ij} = connection weights

5. Training Process

Boltzmann Machine

1. Initialize weights randomly.
2. Input data is given to visible units.
3. Hidden units calculate their states probabilistically.
4. The network samples different configurations.
5. Weights are updated to reduce the difference between the observed and model distributions.
6. Repeat until the model learns the data distribution.

Deep Belief Network

1. Prepare and preprocess the input data.
2. Train the first RBM using the input.

3. Use the learned hidden representation as input to the next RBM.
4. Repeat layer-by-layer (greedy layer-wise pretraining).
5. Fine-tune the complete network using supervised or unsupervised learning.

6. Feature-Learning Capability

Feature	Boltzmann Machine	Deep Belief Network
Structure	Single probabilistic network	Multiple hidden layers
Visible units	Yes	Yes
Hidden units	Yes	Multiple layers
Feature learning	Good	Very good
Learning	Energy-based	Layer-wise + fine-tuning
Complexity	High	High
Representation	Lower-level patterns	Hierarchical features
Typical use	Probability modeling	Recognition and representation learning

7. Real-World Application: Handwritten Digit Recognition

A suitable application is handwritten digit recognition.

For example, a system receives images of handwritten digits from 0–9.

Handwritten Image

↓

Visible Layer

↓

RBM / Hidden Layer 1

↓

RBM / Hidden Layer 2

↓

Higher-Level Features

↓

Digit Classification

↓

Output: "7"

Working:

- The visible layer receives pixel values.
- The first hidden layer learns simple patterns such as edges and curves.
- Deeper layers learn combinations of these patterns, such as digit shapes.
- The learned representation is used to identify the digit.
- This approach can be useful in automatic form processing, postal-code recognition, and document digitization.

8. Advantages

Boltzmann Machines

- Can model complex probability distributions.
- Perform unsupervised feature learning.
- Useful for discovering hidden relationships in data.

Deep Belief Networks

- Learn hierarchical representations.
- Reduce the need for manually designed features.
- Can work effectively with high-dimensional data.
- Useful for classification, recognition, and dimensionality reduction.

9. Limitations

- Training can be computationally expensive.
- Learning may require many iterations.
- Boltzmann Machines can be difficult to train because of their large number of connections.
- DBNs may require careful initialization and hyperparameter selection.
- Modern deep learning architectures such as CNNs and Transformers are often preferred for many large-scale applications.

10. Conclusion

- Boltzmann Machines are probabilistic, energy-based neural networks that learn data distributions using visible and hidden units.

- Deep Belief Networks extend this concept by stacking multiple layers to learn hierarchical features.
- Both are important in the development of deep generative and representation-learning models.
- For applications such as handwritten digit recognition, DBNs can automatically learn useful features from raw image data and use them for accurate recognition.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand